

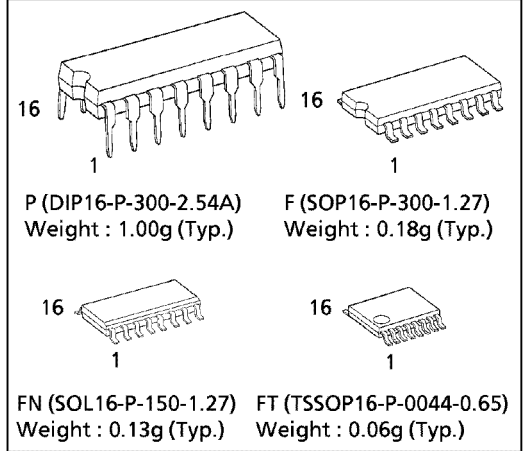
TOSHIBA CMOS DIGITAL INTEGRATED CIRCUIT SILICON MONOLITHIC

TC4051BP, TC4051BF, TC4051BFN, TC4051BFT
TC4052BP, TC4052BF, TC4052BFN, TC4052BFT
TC4053BP, TC4053BF, TC4053BFN, TC4053BFT

TC4051B SINGLE 8-CHANNEL MULTIPLEXER / DEMULTIPLEXER
TC4052B DIFFERENTIAL 4-CHANNEL MULTIPLEXER / DEMULTIPLEXER
TC4053B TRIPLE 2-CHANNEL MULTIPLEXER / DEMULTIPLEXER

(Note) The JEDEC SOP (FN) is not available in Japan.

TC4051B, TC4052B and TC4053B are multiplexers with capabilities of selection and mixture of analog signal and digital signal. TC4051B has 8 channels configuration. TC4052B has 4 channel×2 configuration and TC4053B has 2 channel×3 configuration. The digital signal to the control terminal turns "ON" the corresponding switch of each channel, with large amplitude ($V_{DD}-V_{EE}$) can be switched by the control signal with small logical amplitude ($V_{DD}-V_{SS}$). For example, in the case of $V_{DD}=5V$, $V_{SS}=0V$ and $V_{EE}=-5V$, signals between $-5V$ and $+5V$ can be switched from the logical circuit with single power supply of 5 volts. As the ON-resistance of each switch is low, these can be connected to the circuits with low input impedance.



MAXIMUM RATINGS

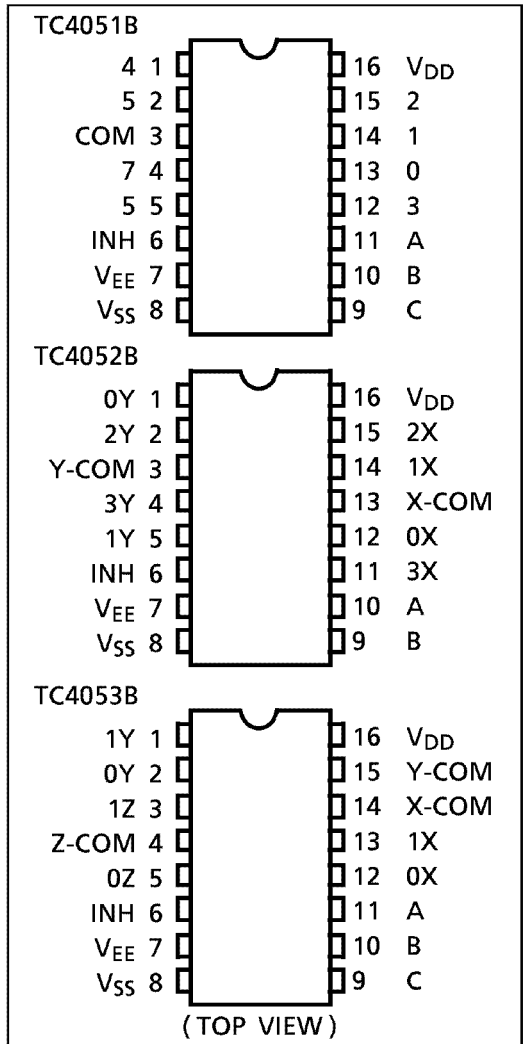
CHARACTERISTIC	SYMBOL	RATING	UNIT
DC Supply Voltage	$V_{DD}-V_{SS}$	-0.5~20	V
DC Supply Voltage	$V_{DD}-V_{EE}$	-0.5~20	V
Control Input Voltage	V_{CIN}	$V_{SS}-0.5\sim V_{DD}+0.5$	V
Switch I/O Voltage	V_I/V_O	$V_{EE}-0.5\sim V_{DD}+0.5$	V
Control Input Current	I_{CIN}	± 10	mA
Potential difference across I/O during ON	V_I-V_O	-0.5~0.5	V
Power Dissipation	P_D	300 (DIP) / 180 (SOIC)	mW
Operating Temperature Range	T_{opr}	-40~85	°C
Storage Temperature Range	T_{stg}	-65~150	°C

TRUTH TABLE

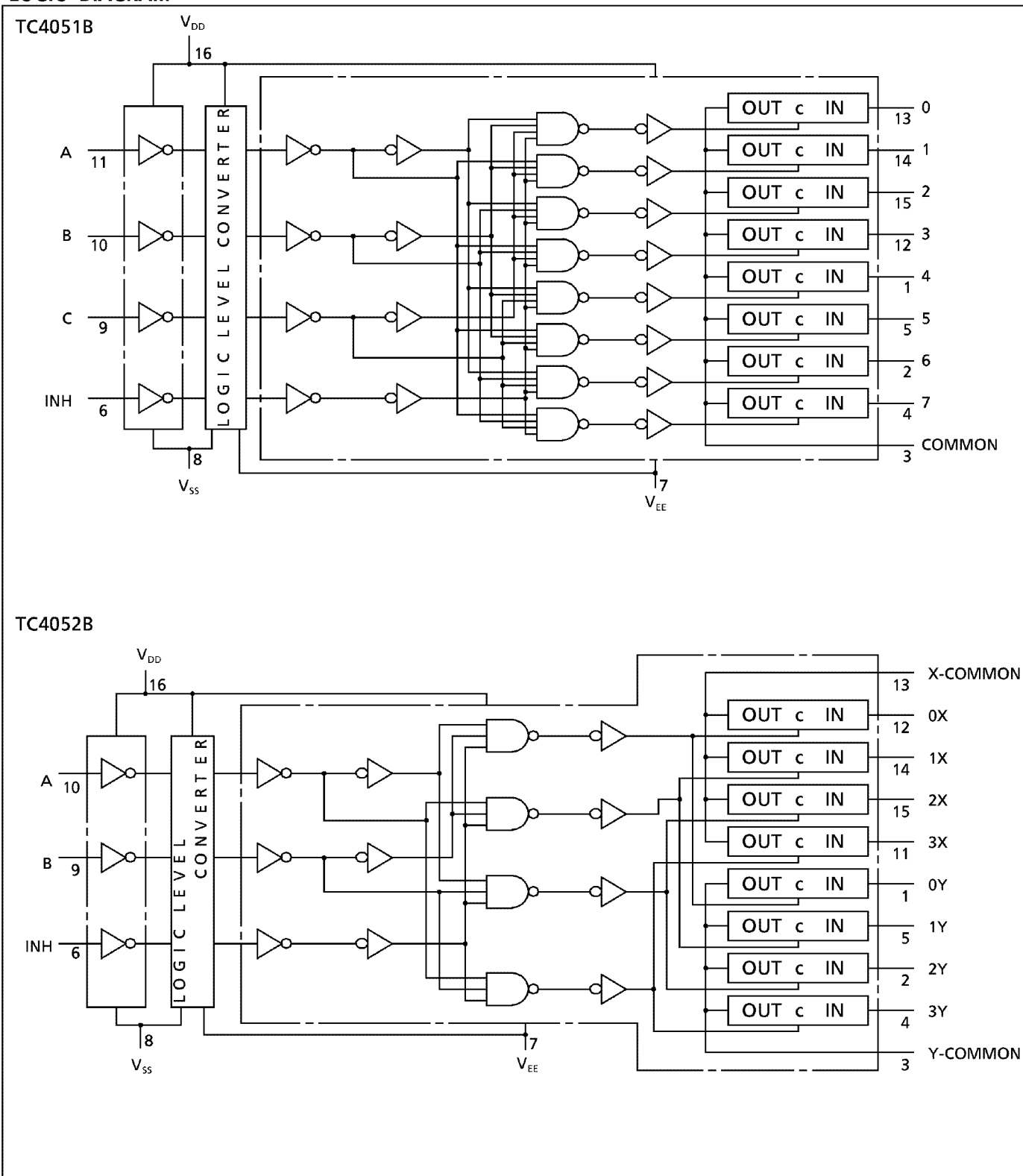
CONTROL INPUTS				"ON" CHANNEL		
INHIBIT	$C\triangle$	B	A	TC4051B	TC4052B	TC4053B
L	L	L	L	0	0X, 0Y	0X, 0Y, 0Z
L	L	L	H	1	1X, 1Y	1X, 0Y, 0Z
L	L	H	L	2	2X, 2Y	0X, 1Y, 0Z
L	L	H	H	3	3X, 3Y	1X, 1Y, 0Z
L	H	L	L	4	—	0X, 0Y, 1Z
L	H	L	H	5	—	1X, 0Y, 1Z
L	H	H	L	6	—	0X, 1Y, 1Z
L	H	H	H	7	—	1X, 1Y, 1Z
H	*	*	*	NONE	NONE	NONE

*: Don't Care $\triangle$ Except TC4052B

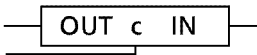
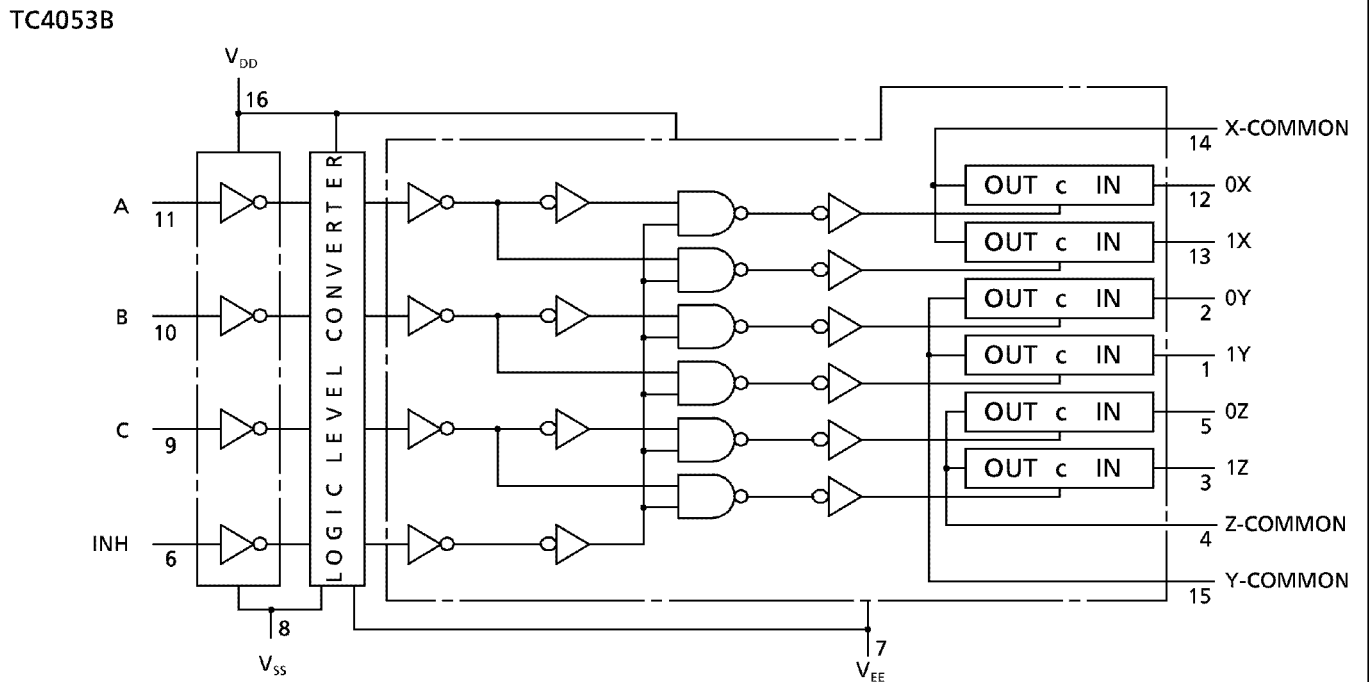
PIN ASSIGNMENT



LOGIC DIAGRAM



LOGIC DIAGRAM



TRUTH TABLE

CONTROL C	Impedance Between IN-OUT*
H	$0.5 \sim 5 \times 10^2 \Omega$
L	$> 10^9 \Omega$

* See Electrical Characteristics

RECOMMENDED OPERATING CONDITIONS

CHARACTERISTIC	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
DC Supply Voltage	$V_{DD}-V_{SS}$		3	—	18	V
	$V_{DD}-V_{EE}$		3	—	18	
Control Input Voltage	V_{IN}		V_{SS}	—	V_{DD}	V
Input / Output Voltage	V_{IN} / V_{OUT}		V_{EE}	—	V_{DD}	V

STATIC ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYM- BOL	TEST CONDI- TION	V _{SS} (V)	V _{EE} (V)	V _{DD} (V)	- 40°C		25°C			85°C		UNIT
						MIN.	MAX.	MIN.	TYP.	MAX.	MIN.	MAX.	
Control Input High Voltage	V _{IH}	V _{IS} = V _{DD} thru 1kΩ	V _{EE} = V _{SS} R _L = 1kΩ to V _{SS} I _{IS} < 2μA on all OFF channels		5	3.5	—	3.5	2.75	—	3.5	—	V
			10	7.0	—	7.0	5.50	—	7.0	—			
			15	11.0	—	11.0	8.25	—	11.0	—			
Control Input Low Voltage	V _{IL}		5	—	1.5	—	2.25	1.5	—	1.5			
			10	—	3.0	—	4.5	3.0	—	3.0			
			15	—	4.0	—	6.75	4.0	—	4.0			
On-State Resistance	R _{ON}	0 ≤ V _{IS} ≤ V _{DD} R _L = 10kΩ	0	0	5	—	850	—	240	950	—	1200	Ω
			0	0	10	—	210	—	110	250	—	300	
			0	0	15	—	140	—	80	160	—	200	
△On-State Resistance Between Any 2 Switches	R _{ON} △		0	0	5	—	—	—	10	—	—	—	
			0	0	10	—	—	—	6	—	—	—	
			0	0	15	—	—	—	4	—	—	—	
Input / Output Leakage Current	I _{OFF}	V _{IN} = 18V, V _{OUT} = 0V V _{IN} = 0V, V _{OUT} = 18V			18 18	— —	± 100 ± 100	— —	± 0.01 ± 0.01	± 100 ± 100	— —	± 1000 ± 1000	nA
Quiescent Supply Current	I _{DD}	V _{IN} = V _{SS} , V _{DD} *			5 10 15	— — —	5.0 10 20	— — —	0.005 0.010 0.015	5.0 10 20	— — —	150 300 600	μA
Input Current	I _{IN}	V _{IH} = 18V V _{IL} = 0V			18 18	— —	0.1 -0.1	— —	10 ⁻⁵ - 10 ⁻⁵	0.1 -0.1	— —	1.0 - 1.0	
Input Capacitance	C _{IN}					—	—	—	5	7.5	—	—	pF
Switch Input Capacitance	C _{IN}					—	—	—	10	—	—	—	
Output Capacitance	C _{OUT}	TC4051B TC4052B TC4053B			10 10 10	— — —	— — —	— — —	58 30 17	— — —	— — —	— — —	
Feedthrough Capacitance	C _{IN- C-OUT}	TC4051B TC4052B TC4053B			10 10 10	— — —	— — —	— — —	0.2 0.2 0.2	— — —	— — —	— — —	

* All valid input combinations.

DYNAMIC ELECTRICAL CHARACTERISTICS (Ta = 25°C, CL = 50pF)

CHARACTERISTIC	SYMBOL	TEST CONDITION	V _{SS} (V) V _{EE} (V) V _{DD} (V)			MIN.	TYP.	MAX.	UNIT
Phase Difference Between Input to Output	$\phi_I - O$		0 0 0	0 0 0	5 10 15	— — —	15 8 6	45 20 15	ns
Propagation Delay Time (A, B, C, - OUT)	t_{pZL}	$R_L = 1k\Omega$	0	0	5	—	170	550	
	t_{pZH}		0	0	10	—	90	240	
	t_{pLZ}		0	0	15	—	70	160	
	t_{pHZ}		0	-5	5	—	100	240	
	t_{pHZ}		0	-7.5	7.5	—	80	160	
Propagation Delay Time (INH - OUT)	t_{pZL}	$R_L = 1k\Omega$	0	0	5	—	120	380	
	t_{pZH}		0	0	10	—	60	200	
	t_{pLZ}		0	0	15	—	50	160	
	t_{pHZ}		0	-5	5	—	80	200	
	t_{pHZ}		0	-7.5	7.5	—	60	160	
Propagation Delay Time (INH - OUT)	t_{pLZ}	$R_L = 1k\Omega$	0	0	5	—	170	450	
	t_{pZH}		0	0	10	—	90	210	
	t_{pLZ}		0	0	15	—	70	160	
	t_{pHZ}		0	-5	5	—	100	210	
	t_{pHZ}		0	-7.5	7.5	—	80	160	
- 3dB Cutoff Frequency TC4051B TC4052B TC4053B	f_{MAX} (I - O)	$R_L = 1k\Omega$ (*1)	-5 -5 -5	-5 -5 -5	5 5 5	— — —	20 30 40	— — —	MHz
Total Harmonic Distortion	—	$R_L = 10k\Omega$ $f = 1kHz$ (*2)	-2.5 -5 -7.5	-2.5 -5 -7.5	2.5 5 7.5	— — —	0.15 0.03 0.02	— — —	%
- 50dB Feedthrough (SWITCH OFF)	—	$R_L = 1k\Omega$ (*3)	-5	-5	5	—	500	—	kHz
Crosstalk	—	$R_L = 1k\Omega$ (*4)	-5	-5	5	—	1.5	—	MHz
Crosstalk (CONTROL - OUT)	—	$R_{IN} = 1k\Omega$ $R_{OUT} = 10k\Omega$ $C_L = 15pF$	0 0 0	0 0 0	5 10 15	— — —	200 400 600	— — —	mV

*1 Sine wave of $\pm 2.5V_{p-p}$ shall be used for V_{is} and the frequency of $20 \log 10 \frac{V_{OS}}{V_{is}} = -3dB$ shall be f_{MAX} .

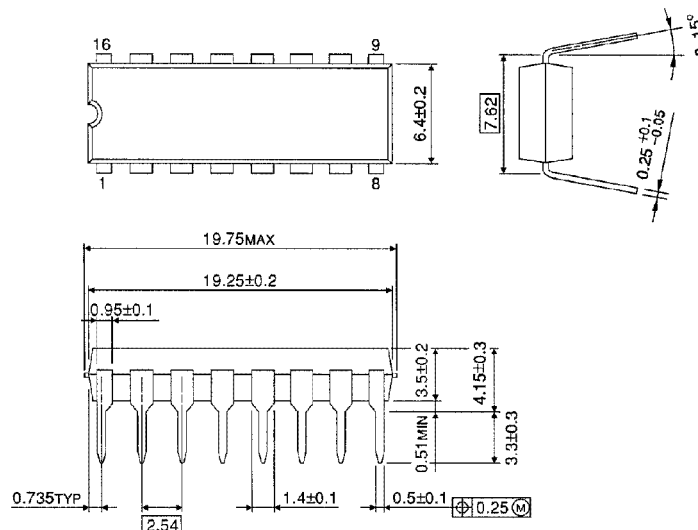
*2 V_{is} shall be sine wave of $\pm \left(\frac{V_{DD} - V_{EE}}{4} \right) p-p$.

*3 Sine wave of $\pm 2.5V_{p-p}$ shall be used for V_{is} and the frequency of $20 \log 10 \frac{V_{OS}}{V_{is}} = -50dB$ shall be feed-through.

*4 Sine wave of $\pm 2.5V_{p-p}$ shall be used for V_{is} and the frequency of $20 \log 10 \frac{V_{OS}}{V_{is}} = -50dB$ shall be Crosstalk.

DIP 16PIN PACKAGE DIMENSIONS (DIP16-P-300-2.54A)

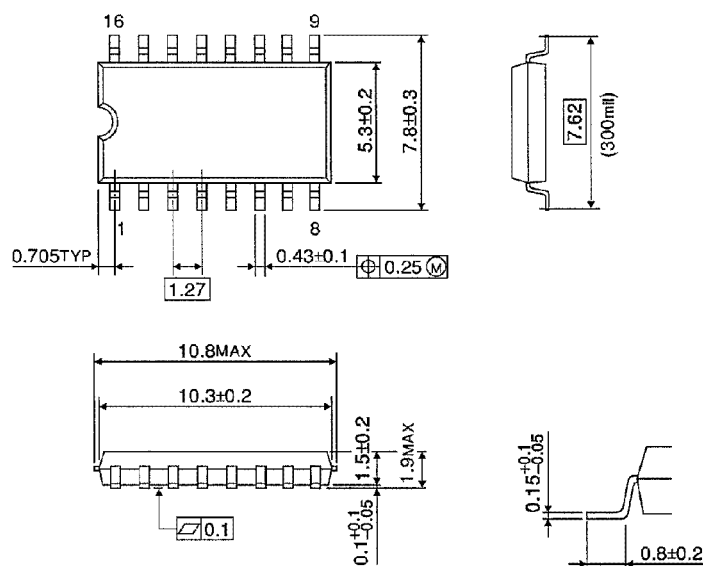
Unit in mm



Weight : 1.00g (Typ.)

SOP 16PIN (200mil BODY) PACKAGE DIMENSIONS (SOP16-P-300-1.27)

Unit in mm

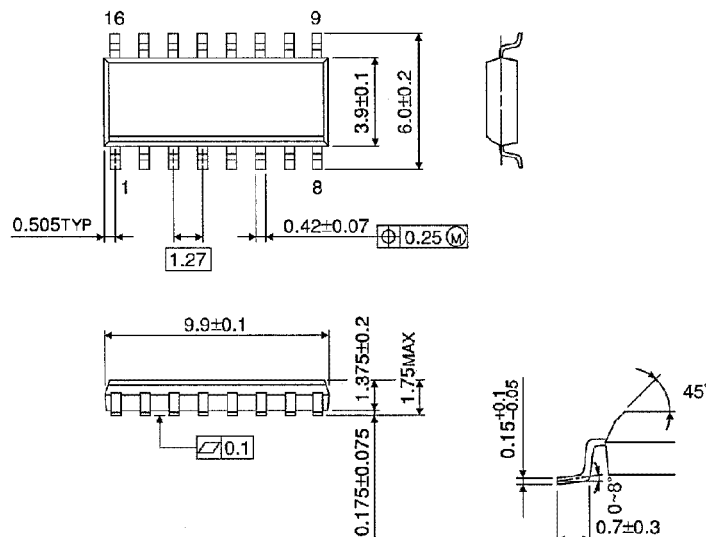


Weight : 0.18g (Typ.)

SOP 16PIN (150mil BODY) PACKAGE DIMENSIONS (SOL16-P-150 -1.27)

Unit in mm

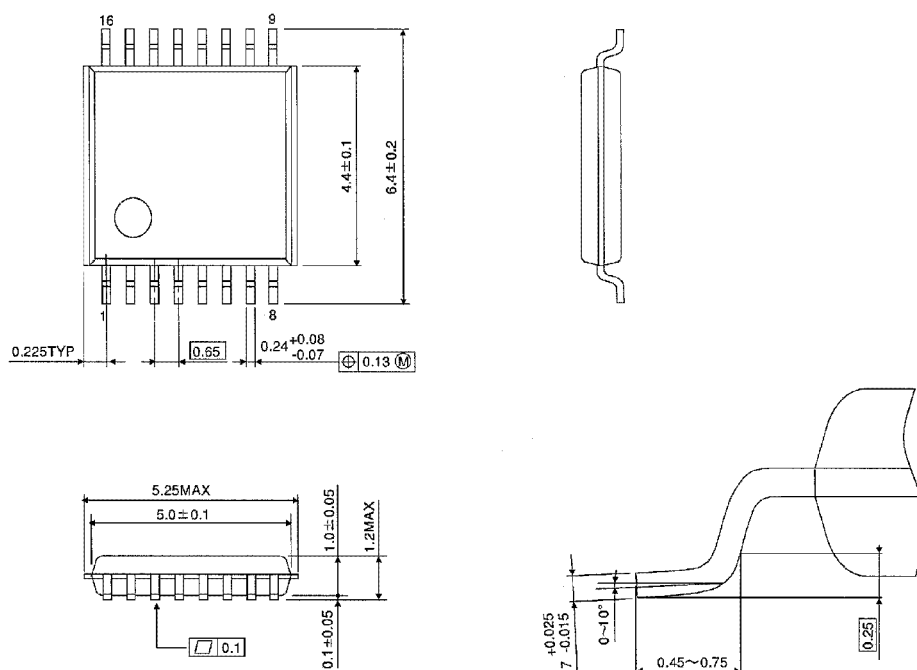
(Note) This package is not available in Japan.



Weight : 0.13g (Typ.)

TSSOP 16PIN PACKAGE DIMENSIONS (TSSOP16-P-0044-0.65)

Unit in mm



Weight : 0.06g (Typ.)

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